Chapter 3

Composition of Substances and Solutions



3.1. Formula Mass and Mole Concept

The masses of atoms and the chemical formulas of compounds allows us to quantitatively describe the composition of substances.

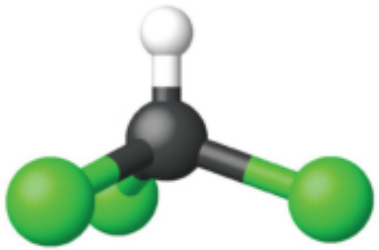
3.1.1. Formula Mass

Formula Mass for Covalent Substances

- For covalent substances, the formula represents the numbers and types of atoms composing a single molecule of the substance;
- The formula mass may be correctly referred to as a molecular mass. Consider chloroform ($CHCl_3$), indicates that a single molecule contains one carbon atom, one hydrogen atom, and three chlorine atoms.

Eg.1 The average molecular mass of a chloroform molecule is therefore equal to the sum of the average atomic masses of these atoms.

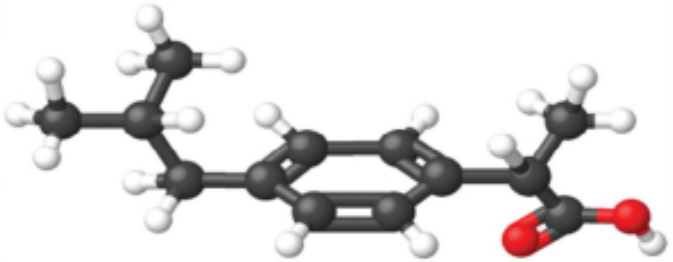
Element	Quantity		Average atomic mass (amu)		Subtotal (amu)
C	1	×	12.01	=	12.01
H	1	×	1.008	=	1.008
Cl	3	×	35.45	=	106.35
				Molecular mass	119.37



Eg.2 Ibuprofen, $C_{13}H_{18}O_2$, is a covalent compound and the active ingredient in several popular nonprescription pain medications, such as Advil and Motrin.

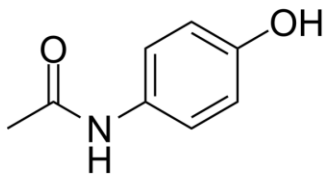
What is the molecular mass (amu) for this compound?

Element	Quantity		Average atomic mass (amu)		Subtotal (amu)
C	13	×	12.01	=	156.13
H	18	×	1.008	=	18.144
O	2	×	16.00	=	32.00
				Molecular mass	206.27



Exercise

Acetaminophen, $C_8H_9NO_2$, is a covalent compound and the active ingredient in several popular nonprescription pain medications, such as Tylenol. What is the molecular mass (amu) for this compound?

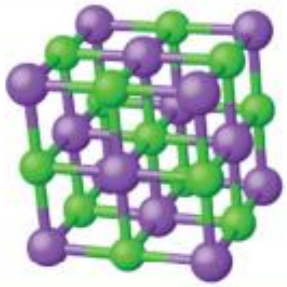


Formula Mass of Ionic Compounds

- Ionic compounds are composed of discrete cations and anions combined in ratios to yield electrically neutral bulk matter.
- The formula mass for an ionic compound is calculated in the same way as the formula mass for covalent compounds: by summing the average atomic masses of all the atoms in the compound's formula.
- The formula for an ionic compound does not represent the composition of a discrete molecule, so it may not correctly be referred to as the "molecular mass."

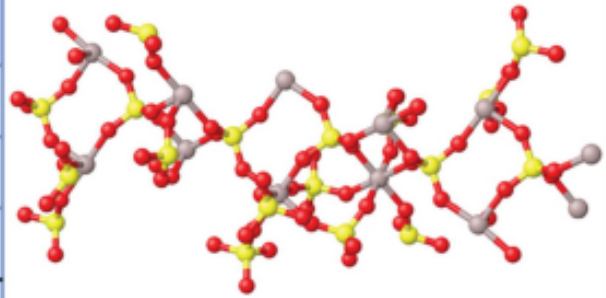
Eg.1 Consider sodium chloride, NaCl, the chemical name for common table salt.

Element	Quantity		Average atomic mass (amu)		Subtotal
Na	1	×	22.99	=	22.99
Cl	1	×	35.45	=	35.45
				Formula mass	58.44



Eg.2 Aluminum sulfate, $Al_2(SO_4)_3$, is an ionic compound that is used in the manufacture of paper and in various water purification processes. What is the formula mass (amu) of this compound?

Element	Quantity		Average atomic mass (amu)		Subtotal (amu)
Al	2	×	26.98	=	53.96
S	3	×	32.06	=	96.18
O	12	×	16.00	=	192.00
				Molecular mass	342.14



Exercise

Calcium phosphate, $Ca_3(PO_4)_2$, is an ionic compound and a common anti-caking agent added to food products. What is the formula mass (amu) of calcium phosphate?

3.1.2 The Mole Concept

- The identity of a substance is defined not only by the types of atoms or ions it contains, but by the quantity of each type of atom or ion.
  For example, water, H_2O, and hydrogen peroxide, H_2O_2, are alike in that their respective molecules are composed of hydrogen and oxygen atoms.
- The mole is an amount unit similar to familiar units like pair, dozen, gross, etc.
- A mole of sul … screte entities (ator …
- This large nu … (NA) or the Avogadr … constant is properly rep … ed version being 6.022 …
- 1 mole of any … other element.
- However, the … masses of the individual ato …

The Mole

1 dozen = 12

1 gross = 144

1 ream = 500

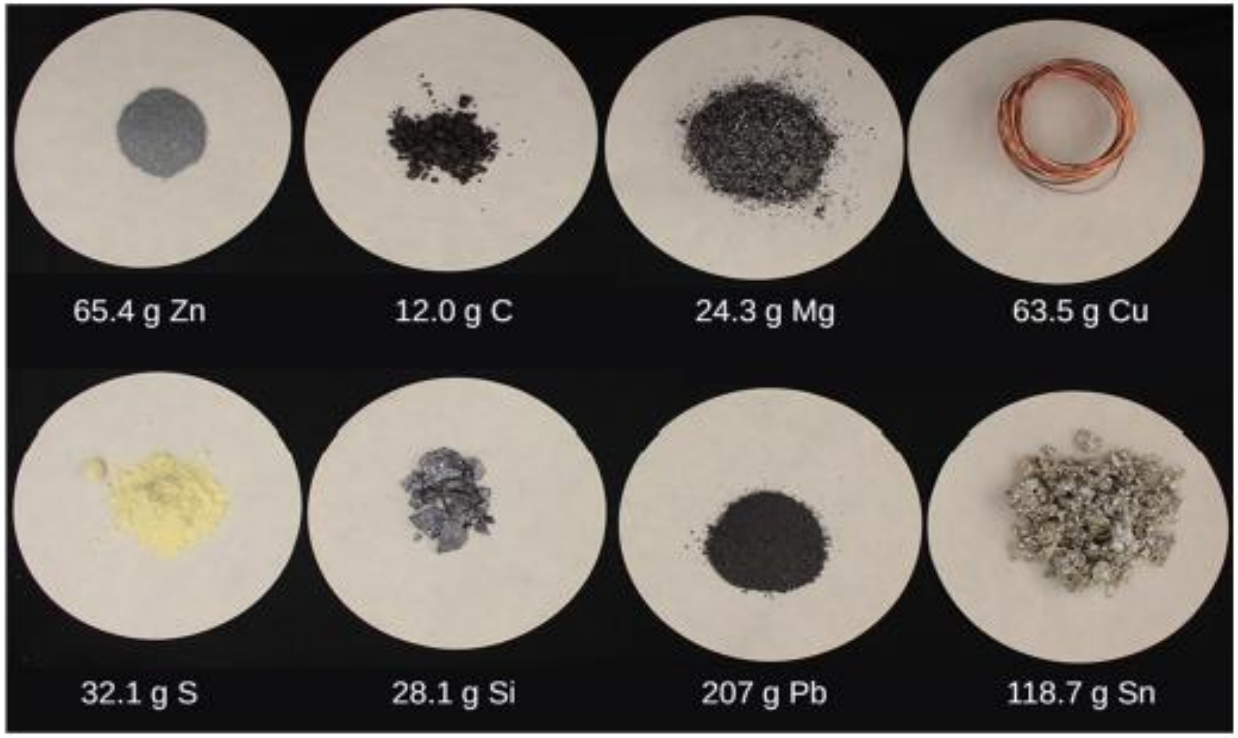
1 mole = 6.02×10^{23}



There are exactly 12 grams of carbon-12 in one mole of carbon-12.

Mole

- The molar mass of an element (or compound) is the **mass in grams of 1 mole of that substance**, a property expressed in units of grams per mole (g/mol)



Each sample contains 6.022 X 10^{23} atoms = 1.00 mol of atoms.

- The molar mass of any substance is numerically equivalent to its atomic or formula weight in amu. Per the amu definition, a single ^{12}C atom weighs 12 amu (its atomic mass is 12 amu). A mole of ^{12}C weighs 12 g (its molar mass is 12 g/mol).

- The molar mass of a compound in grams is likewise numerically equivalent to its formula mass in amu.

Example: According to nutritional guidelines from the US Department of Agriculture, the estimated average requirement for dietary potassium is 4.7 g. What is the n estimated average requirement of potassium in moles?

Solution

$$4.7 \cancel{\text{g}}\ \text{K}\left(\frac{\text{mol K}}{39.10 \cancel{\text{g}}}\right) = 0.12\ \text{mol K}$$

3.2 Determining Empirical and Molecular Formulas

- Given the chemical formula of the substance, one may determine the amount of the substance (moles) from its mass, and vice versa.

But what if the chemical formula of a substance is unknown?

- In this section, we will explore how to apply these very same principles in order to derive the chemical formulas of unknown substances from experimental mass measurements.

Percent Composition

- The elemental makeup of a compound defines its chemical identity, and chemical formulas are the most succinct (concise) way of representing this elemental makeup.
- When a compound's formula is unknown, measuring the mass of each of its constituent elements is often the first step in the process of determining the formula experimentally.
- The results of these measurements permit the calculation of the compound's percent composition, defined as the percentage by mass of each element in the compound.

Percent Composition

- For example, consider a gaseous compound composed solely of carbon and hydrogen. The percent composition of this compound could be represented as follows:

$$\%\,\mathrm{H} = \frac{\text{mass H}}{\text{mass compound}} \times 100\%$$

$$\%\,\mathrm{C} = \frac{\text{mass C}}{\text{mass compound}} \times 100\%$$

If analysis of a 10.0-g sample of this gas showed it to contain 2.5 g H and 7.5 g C, the percent composition would be calculated to be 25% H and 75% C:

$$\%\,\mathrm{H} = \frac{2.5\text{ g H}}{10.0\text{ g compound}} \times 100\% = 25\%$$

$$\%\,\mathrm{C} = \frac{7.5\text{ g C}}{10.0\text{ g compound}} \times 100\% = 75\%$$

Example: Analysis of a 12.04-g sample of a liquid compound composed of carbon, hydrogen, and nitrogen showed it to contain 7.34 g C, 1.85 g H, and 2.85 g N. What is the percent composition of this compound?

Percent Composition

Solution

To calculate percent composition, divide the experimentally derived mass of each element by the overall mass of the compound, and then convert to a percentage:

$$\%H = \frac{1.85gH}{12.04g\ compound}\ X\ 100\% = 15.4\%$$

$$\%N = \frac{2.85gN}{12.04g\ compound}\ X\ 100\% = 23.7\%$$

$$\%C = \frac{7.34gC}{12.04g\ compound}\ X\ 100\% = 61.0\%$$

The analysis results indicate that the compound is 61.0% C, 15.4% H, and 23.7% N by mass.

Determining Percent Composition from Formula Mass

- The percent composition of a compound is easily derived from its formula mass and the atomic masses of its constituent elements.
- A molecule of NH_3 contains one N atom weighing 14.01 amu and three H atoms weighing a total of (3 × 1.008 amu) = 3.024 amu.
- The formula mass of ammonia is therefore (14.01 amu + 3.024 amu) = 17.03 amu, and its percent composition is:

$$\%\,\mathrm{N} = \frac{14.01\ \text{amu N}}{17.03\ \text{amu NH}_3} \times 100\% = 82.27\%$$

$$\%\,\mathrm{H} = \frac{3.024\ \text{amu N}}{17.03\ \text{amu NH}_3} \times 100\% = 17.76\%$$

Determination of Empirical Formulas

- Chemical formulas represent the relative numbers, not masses, of atoms in the substance.
- Consider a sample of compound determined to contain 1.71 g C and 0.287 g H. The corresponding numbers of atoms (in moles) are:

$$1.17\ \text{g C} \times \frac{1\ \text{mol C}}{12.01\ \text{g C}} = 0.142\ \text{mol C}$$

$$0.287\ \text{g H} \times \frac{1\ \text{mol H}}{1.008\ \text{g H}} = 0.284\ \text{mol H}$$

- Thus, we can accurately represent this compound with the formula $\mathbf{C_{0.142}H_{0.248}}$.
- Formulas contain whole-number subscripts, which can be achieved by dividing each subscript by the smaller subscript:

$$\text{C}\ \frac{0.142}{0.142}\ \ \text{H}\ \frac{0.248}{0.142}\ \ \text{or}\ \text{CH}_2$$

- The empirical formula for this compound is thus CH2.
- This may or not be the compound's molecular formula as well

Determination of Empirical Formulas

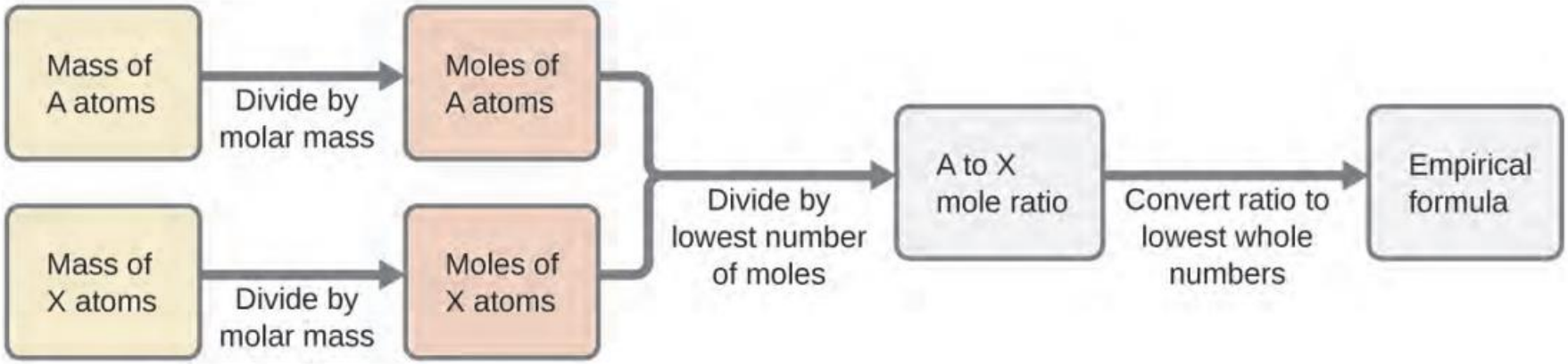
Example: a sample of compound determined to contain 5.31 g Cl and 8.40 g O. Following the same approach yields a tentative empirical formula of:

$$Cl_{0.150}O_{0.525} = Cl_{\frac{0.150}{0.150}}\ O_{\frac{0.525}{0.150}} = ClO_{3.5}$$

Determination of Empirical Formulas

- In summary, empirical formulas are derived from experimentally measured element masses by:

1. Deriving the number of moles of each element from its mass
2. Dividing each element's molar amount by the smallest molar amount to yield subscripts for a tentative empirical formula
3. Multiplying all coefficients by an integer, if necessary, to ensure that the smallest whole-number ratio of subscripts is obtained



Example 1: A sample of the black mineral hematite an oxide of iron found in many iron ores, contains 34.97 g of iron and 15.03 g of oxygen. What is the empirical formula of hematite

Solution

For this problem, we are given the mass in grams of each element. Begin by finding the moles of each:

$$34.97\ \text{g Fe}\left(\frac{\text{mol Fe}}{55.85\ \text{g}}\right) = 0.6261\ \text{mol Fe}$$

$$15.03\ \text{g O}\left(\frac{\text{mol O}}{16.00\ \text{g}}\right) = 0.9394\ \text{mol O}$$

Next, derive the iron-to-oxygen molar ratio by dividing by the lesser number of moles:

$$\frac{0.6261}{0.6261} = 1.000\ \text{mol Fe}$$

$$\frac{0.0394}{0.6261} = 1.500\ \text{mol O}$$

The ratio is 1.000 mol of iron to 1.500 mol of oxygen ($Fe_1O_{1.5}$). Finally, multiply the ratio by two to get the smallest possible whole number subscripts while still maintaining the correct iron-to-oxygen ratio:

$2\ (Fe_1\ O_{1.5)} = Fe_2\ O_3$

The empirical formula is Fe_2O_3.

Derivation of Molecular Formulas

- Molecular formulas are derived by comparing the compound's molecular or molar mass to its empirical formula mass.
- an empirical formula mass is the sum of the average atomic masses of all the atoms represented in an empirical formula.
- If we know the molecular (or molar) mass of the substance, we can divide this by the empirical formula mass in order to identify the number of empirical formula units per molecule, which we designate as n:

$$\frac{\text{molecular or molar mass}\left(\text{amu or }\frac{\text{g}}{\text{mol}}\right)}{\text{empirical formula mass}\left(\text{amu or }\frac{\text{g}}{\text{mol}}\right)} = n \text{ formula units/molecule}$$

The molecular formula is then obtained by multiplying each subscript in the empirical formula by *n*, as shown by the generic empirical formula A_xB_y:

$$(A_x B_y)_n = A_{nx} B_{nx}$$

Example 1: consider a covalent compound whose empirical formula is determined to be CH_2O.

- The empirical formula mass for this compound is approximately 30 amu
- If the compound's molecular mass is determined to be 180 amu, this indicates that molecules of this compound contain six times the number of atoms represented in the empirical formula:

$$\frac{180 \text{ amu/molecule}}{30 \frac{\text{amu}}{\text{formula unit}}} = 6 \text{ formula units/molecule}$$

Molecules of this compound are then represented by molecular formulas whose subscripts are six times greater than those in the empirical formula:

$$(CH_2O)_6 = C_6H_{12}O_6$$

Example 2: Nicotine, an alkaloid produced by plants that is mainly responsible for the addictive nature of cigarettes, contains 74.02% C, 8.710% H, and 17.27% N. If 40.57 g of nicotine contains 0.2500 mol nicotine, what is the molecular formula?

Solution: Determining the molecular formula from the provided data will require comparison of the compound's empirical formula mass to its molar mass.

Derivation of Molecular Formulas

$$(74.02\ \text{g C})\left(\frac{1\ \text{mol C}}{12.01\ \text{g C}}\right) = 6.163\ \text{mol C}$$

$$(8.710\ \text{g H})\left(\frac{1\ \text{mol H}}{1.01\ \text{g H}}\right) = 8.624\ \text{mol H}$$

$$(17.27\ \text{g N})\left(\frac{1\ \text{mol N}}{14.01\ \text{g N}}\right) = 1.233\ \text{mol N}$$

After calculation of the molar ratios of these elements the empirical formula is C_5H_7N.

The molar mass for nicotine from the given mass and molar amount of compound:

$$\frac{40.57\ \text{g nicotine}}{0.2500\ \text{mol nicotine}} = \frac{162.3\ \text{g}}{\text{mol}}$$

- Comparing the molar mass and empirical formula mass indicates that each nicotine molecule contains two formula units:

$$\frac{162.3\ \text{g/mol}}{81.13\ \frac{\text{g}}{\text{formula unit}}} = 2\ \text{formula units/molecule}$$

$$\mathbf{(C_5H_7N)_2 = C_{10}H_{14}N_2}$$

3.3 Molarity

- Mixtures are more commonly encountered in nature than are pure substances.
- the relative composition of a mixture plays an important role in determining its properties. Eg. Alloy,

Solutions

- The relative amount of a given solution component is known as its concentration.
- Often a solution contains one component with a concentration that is significantly greater than that of all other components.
- This component is called the solvent and may be viewed as the medium in which the other components are dispersed, or dissolved.
- A solute is a component of a solution that is typically present at a much lower concentration than the solvent.
- Solute concentrations are often described with qualitative terms such as dilute (of relatively low concentration) and concentrated (of relatively high concentration).

3.3 Molarity and Other Concentration Units

3.3.1 Molarity

- Concentrations may be quantitatively assessed using a wide variety of measurement units, each convenient for particular applications.
- Molarity (M) is a useful concentration unit for many applications in chemistry. Molarity is defined as the **number of moles of solute in exactly 1 liter** (1 L) of the solution:

$$M = \frac{\text{mol solute}}{\text{L solution}}$$

Example: Calculating Molar Concentrations

A 355-mL soft drink sample contains 0.133 mol of sucrose (table sugar). What is the molar concentration of sucrose in the beverage?

Solution

Since the molar amount of solute and the volume of solution are both given, the molarity can be calculated using the definition of molarity. Per this definition, the solution volume must be converted from mL to L:

$$M = \frac{\text{mol solute}}{\text{L solution}} = \frac{0.133\ \text{mol}}{355\ \text{mL} \times \frac{1\ \text{L}}{1000\ \text{mL}}} = 0.375\ M$$

3.3.2. Dilution of Solutions

- Dilution is the process whereby the concentration of a solution is lessened by the addition of solvent.
- Dilution is also a common means of preparing solutions of a desired concentration.
- A simple mathematical relationship can be used to relate the volumes and concentrations of a solution before and after the dilution process.
- According to the definition of molarity, the molar amount of solute in a solution is equal to the product of the solution's molarity and its volume in liters:

$$n = ML$$

Expressions like these may be written for a solution before and after it is diluted:

$$n_1 = M_1L_1 \qquad n_2 = M_2L_2$$

The subscripts "1" and "2" refer to the solution before and after the dilution, respectively

Since the dilution process does not change the amount of solute in the solution, $n_1 = n_2$.

$$\mathbf{C_1V_1 = C_2V_2}$$

Example 1: Determining the Concentration of a Diluted Solution

If 0.850 L of a 5.00-M solution of copper nitrate, $Cu(NO_3)_2$, is diluted to a volume of 1.80 L by the addition of water, what is the molarity of the diluted solution?

Solution: The stock concentration, C_1, and volume, V_1, are provided as well as the volume of the diluted solution, V_2. Rearrange the dilution equation to isolate the unknown property, the concentration of the diluted solution, C_2:

$$C_1V_1=C_2V_2$$

$$C_2=C_1V_1/\ V_2$$

$$C2 = \frac{0.850\ x\ 5.00\ mol}{1.80L} = 2.36\ M$$

Example 2: Volume of a Diluted Solution

What volume of 0.12 M HBr can be prepared from 11 mL (0.011 L) of 0.45 M HBr?

Solution We are given the **volume and concentration of a stock solution**, V_1 and C_1, and the **concentration of the resultant diluted solution**, C_2. We need to find the **volume of the diluted solution**, V_2. We thus rearrange the dilution equation in order to isolate V_2:

$$\mathbf{C_1V_1 = C_2V_2}$$

$$V2 = \frac{C1V1}{C2}$$

$$V2 = \frac{(0.45\ M)(0.011L)}{(0.12M)}$$

V2 = 0.041

3.3.3. Percentage (W/W, W/V and V/V)

3.3.3.1. Mass Percentage

- The mass percentage of a solution component is defined as the ratio of the component's mass to the solution's mass, expressed as a percentage:

$$Mass\ Percentage = \frac{mass\ of\ component}{mass\ of\ solution}\ X\ 100\%$$

- Mass percentage is also referred to by similar names such as percent mass, percent weight, weight/weight percent, and other variations on this theme.

- The most common symbol for mass percentage is simply the percent sign, %, although more detailed symbols are often used including %mass, %weight, and (w/w)%.

- Mass percentages are popular concentration units for consumer products.

Example: A liquid bleach (an aqueous solution of sodium hypochlorite (NaOCl))bottle cites the concentration of its active ingredient (sodium hypochlorite (NaOCl)), as being 7.4%.
A 100.0-g sample of bleach would therefore contain 7.4 g of NaOCl.

Example Calculation of Percent by Mass

A 5.0-g sample of spinal fluid contains 3.75 mg (0.00375 g) of glucose. What is the percent by mass of glucose in spinal fluid?

Solution The spinal fluid sample contains roughly 4 mg of glucose in 5000 mg of fluid, so the mass fraction of glucose should be a bit less than one part in 1000, or about 0.1%.

Substituting the given masses into the equation defining mass percentage yields:

$$\%\ glucose = \frac{3.75mg\ glucose \frac{1g}{1000mg}}{5.0g\ spinal\ fluid} = 0.075\%$$

3.3.3.2. Volume Percentage

❑The concentration of a solution formed by dissolving a liquid solute in a liquid solvent is therefore often expressed as a volume percentage, %vol or (v/v)%:

$$\text{Volume percentage} = \frac{\text{Volume solute}}{Volume\ of\ solution}\ X\ 100\%$$

Example: Calculations using Volume Percentage

Rubbing alcohol (isopropanol) is usually sold as a 70%vol aqueous solution. If the density of isopropyl alcohol is 0.785 g/mL, how many grams of isopropyl alcohol are present in a 355 mL bottle of rubbing alcohol?

Solution

Per the definition of volume percentage, the isopropanol volume is 70% of the total solution volume. Multiplying the isopropanol volume by its density yields the requested mass:

$$(355ml\ solution)\left(\frac{70ml\ isopropyl\ acohol}{100ml\ solution}\right)\left(\frac{0.785g\ isopropyl\ alcohol}{1mL\ isopropyl\ alcohol}\right)$$
$$= 195g\ isopropyl\ alcohol$$

3.3.3.3. Mass-Volume Percentage

- A mass-volume percent is a ratio of a solute's mass to the solution's volume expressed as a percentage.

Example 1.

physiological saline solution has a concentration of 0.9% mass/volume (m/v), indicating that the composition is 0.9 g of solute per 100 mL of solution

Example 2: The NaCl concentration of physiological saline is 0.9% (m/v).

Example 3: The normal range for glucose concentration in blood (fasting) is around 70–100 mg/dL.

3.3.4. Parts per million (ppm) and Part per billion (ppb)

- Very low solute concentrations are often expressed using appropriately small units such as parts per million (ppm) or parts per billion (ppb).
- Like percentage ("part per hundred") units, ppm and ppb may be defined in terms of masses, volumes, or mixed mass-volume units.
- There are also ppm and ppb units defined with respect to numbers of atoms and molecules.

The mass-based definitions of ppm and ppb are given here:

$$ppm \quad \frac{mass\ solute}{mass\ solution} X10^{6} ppm$$

$$ppb = \frac{mass\ solute}{mass\ solution} X10^{9} ppb$$

Example: Calculation of Parts per Million and Parts per Billion Concentrations

According to the EPA, when the concentration of lead in tap water reaches 15 ppb, certain remedial actions must be taken. What is this concentration in ppm? At this concentration, what mass of lead (µg) would be contained in a typical glass of water (300 mL)?

Solution

The definitions of the ppm and ppb units may be used to convert the given concentration from ppb to ppm. Comparing these two unit definitions shows that ppm is 1000 times greater than ppb (1 ppm = 10^3 ppb). Thus:

$$15ppb\ X \frac{1ppm}{10^3 ppb} = 0.015ppm$$

$$ppb = \frac{mass\ solute}{mass\ solution} X10^9 ppb$$

$$mass\ of\ solute = \frac{ppb\ X\ mass\ solution}{10^9 ppb}$$

$$mass\ solute = \frac{15ppb\ X\ 300mL\ X\ \frac{1.00g}{ml}}{10^9 ppb} = 4.5\ X\ 10^{-6} g$$

Finally, convert this mass to the requested unit of micrograms:

$$4.4\ X\ 10^{-6} g\ X\ \frac{1\mu g}{10^{-6} g} = 4.5\mu g$$